

3.2.10	Alcohols	
	Nomenclature	be able to apply IUPAC rules for nomenclature to alcohols, aldehydes, ketones and carboxylic acids limited to chains with up to 6 carbon atoms
	Ethanol production	know how ethanol is produced industrially by fermentation
		know the conditions for this reaction and understand the economic and environmental advantages and disadvantages of this process compared with the industrial production from ethene
		understand the meaning of the term biofuel
		know that the term carbon neutral refers to 'an activity that has no net annual carbon (greenhouse gas) emissions to the atmosphere'
		appreciate the extent to which ethanol, produced by fermentation, can be considered to be a carbon-neutral biofuel
	Classification and reactions	understand that alcohols can be classified as primary, secondary or tertiary
		understand that tertiary alcohols are not easily oxidised
		understand that primary alcohols can be oxidised to aldehydes and carboxylic acids and that secondary alcohols can be oxidised to ketones by a suitable oxidising agent such as acidified potassium dichromate(VI) (equations showing [O] as oxidant are acceptable)
		be able to use a simple chemical test to distinguish between aldehydes and ketones (e.g. Fehling's solution or Tollens' reagent)
	Elimination	know that alkenes can be formed from alcohols by acid catalysed elimination reactions (mechanism not required)
		appreciate that this method provides a possible route to polymers without using monomers derived from oil